1.64 Graphs of Motion

Worked Solutions

1 a)

2 metres from the start of the track

1 b)

4 metres from the start of the track

1 c)

$\frac{4-2}{8-0} = 0.25$ ms^{-1}

1 d)

$0.25 \times 5 + 2 = 3.25$ metres

(1.25 metres from where the car started)

2 a)

$16 + 17 + 7 = 40$ m

2 b)

-8 m

2 c)

$\frac{40}{20} = 2$ ms^{-1}

2 d)

$\frac{16}{4} = 4$ ms^{-1}

2 e)

$-\frac{24}{16} = -\frac{3}{2}$ ms^{-1}

2 f)

$-\frac{8}{20} = -\frac{2}{5}$ ms^{-1}

3 a)

$6 + 12 = 18$ m

3 b)

6 m

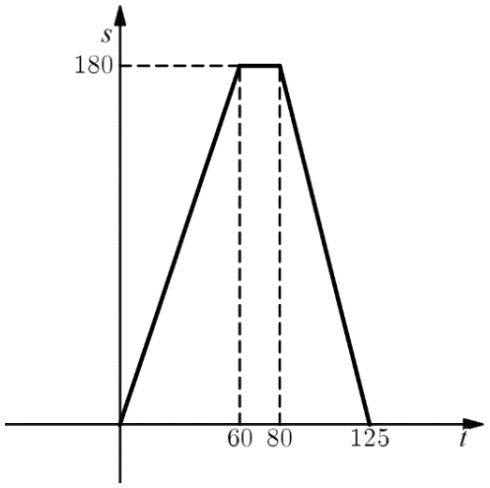
3 c)

$\frac{18}{10} = 1.8$ ms^{-1}

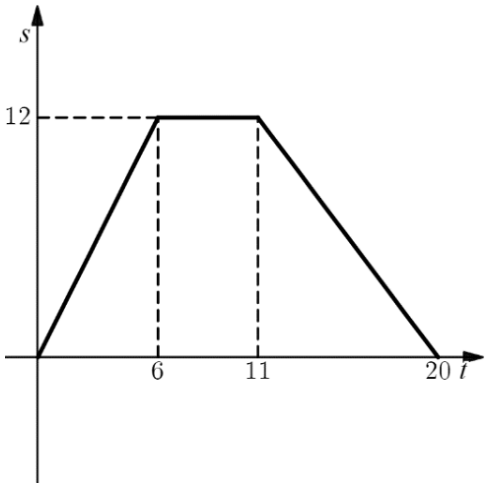
3 d)

$\frac{6}{10} = 0.6 \text{ ms}^{-1}$

4)



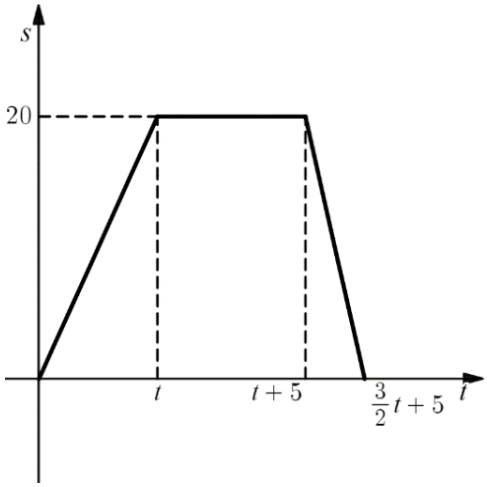
5 a)



5 b)

$\frac{12-0}{11-20} = -\frac{4}{3} \text{ ms}^{-1}$

6 a)



6 b)

Stationary for 5 seconds, so $\frac{3}{2}t + 5 = 11$

$\therefore t = 4$ seconds

7 a)

The object starts at 2 ms^{-1} and it accelerates uniformly until it reaches 4 ms^{-1}

It remains at a constant speed until a total of 6 seconds have passed since the start.

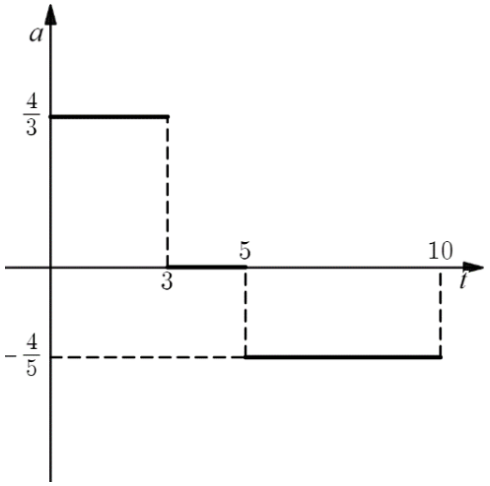
7 b)

The object starts from rest, accelerating at 2 ms^{-2} for the first 3 seconds and reaches 6 ms^{-1}.

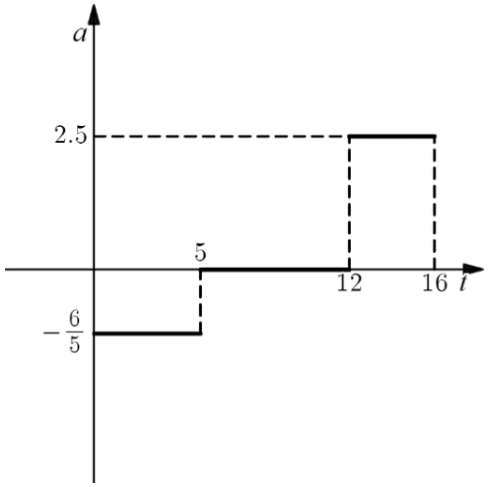
The object travels at 6 ms^{-1} for 4 seconds.

The object decelerates at 4 ms^{-2} uniformly, before reaching a speed of 0 ms^{-1} in 1.5 seconds. It then accelerates in the negative direction at 4 ms^{-2} for a further 1.5 seconds.

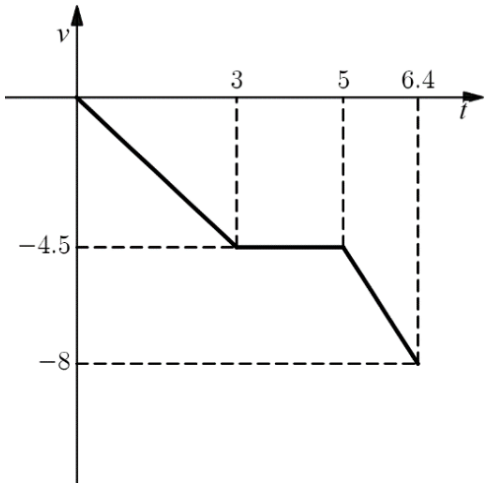
8 a)



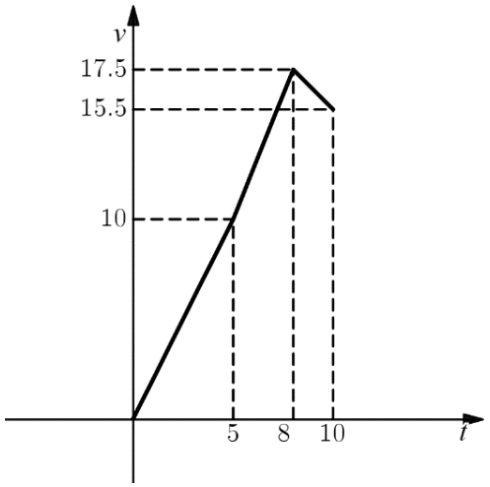
8 b)



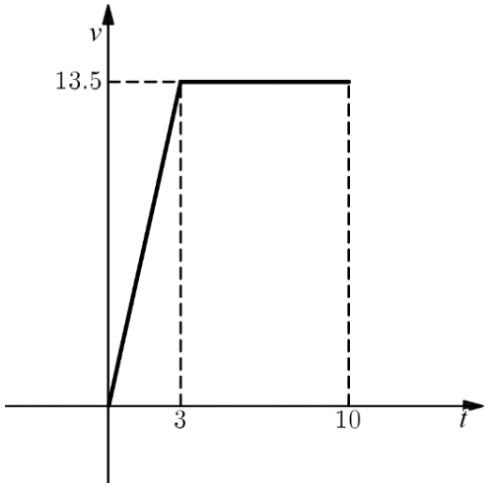
9 a)



9 b)



10 a)



10 b)

$\frac{1}{2} \times 3 \times 13.5 + 7 \times 13.5 = 114.75$ m

11 a)

0.5 ms^{-2}, -0.2 ms^{-2}, $\frac{2}{3}$ ms^{-2}

11 b)

$\frac{1}{2} \times 5 \times 10 + \frac{1}{2}(5 + 4.2) \times 4 + \frac{1}{2}(4.2 + 8.2) \times 6$

$= 80.6$ m

11 c)

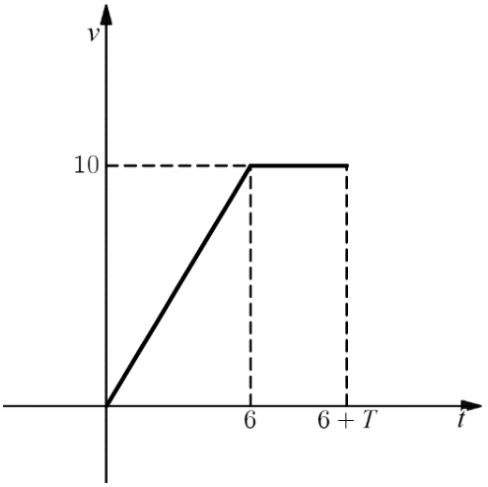
On the line $v = 0.5t$ so if $v = 4$, $t = 8$ s

11 d)

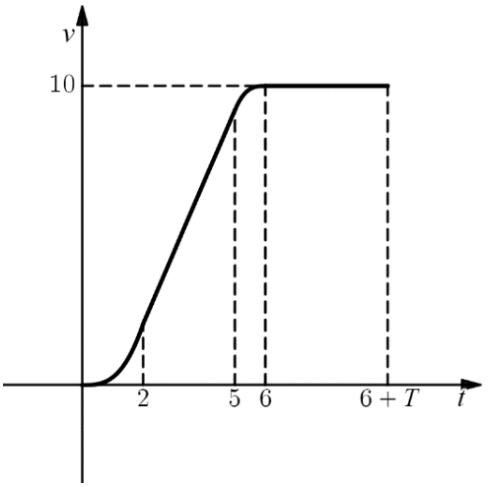
On the line $v - 4.2 = \frac{2}{3}(t - 14)$

So if $t = 17$, $v = 6.2$ ms^{-1}

12 a)



12 b)



13 a)

$-\frac{6}{10} = -0.6$ ms^{-2}

So deceleration is 0.6 ms^{-2}

13 b)

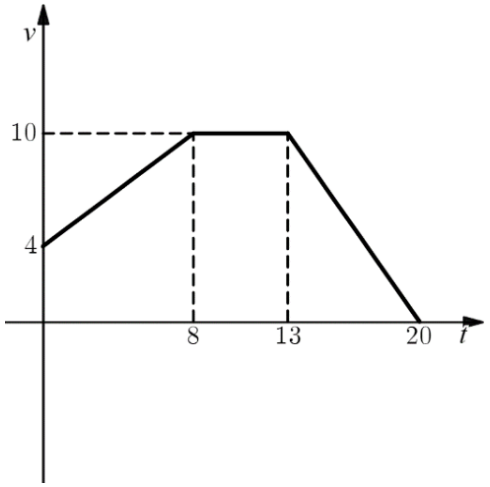
$\frac{1026}{18} = 57$ s

13 c)

$\frac{1}{2}(24 + 18) \times 10 = 210$

$1026 + 210 = 1236$ m

14 a)



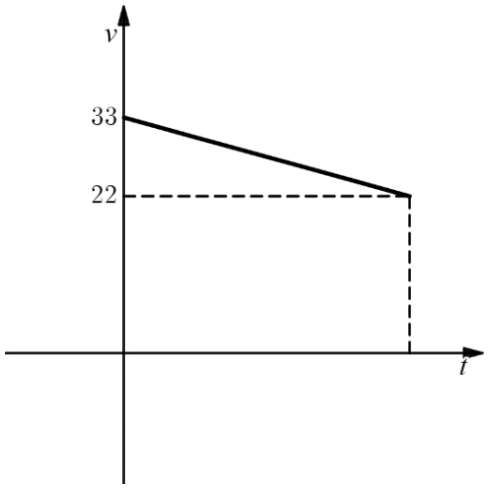
14 b)

$\frac{1}{2}(4+10)\times 8+5\times 10+\frac{1}{2}\times 7\times 10=141$ m

14 c)

$\frac{141}{20}=7.05$ ms^{-1}

15 a)



15 b)

$1100=\frac{1}{2}(33+22)\times t$ so $t=40$ seconds

15 c)

$22\times T=400$ so $T=\frac{200}{11}$

So total time is $40+\frac{200}{11}=\frac{640}{11}$

Average speed is $1500\div\frac{640}{11}=\frac{825}{32}=25.8$ ms^{-1}

15 d)

$\frac{1}{2} \times (33 + 22) = 27.5 \text{ ms}^{-1}$

16 a)

$\frac{6}{5} = 1.2 \text{ ms}^{-2}$, 0 ms^{-2}, $-\frac{11}{22} = -0.5 \text{ ms}^{-2}$, $\frac{5}{5} = 1 \text{ ms}^{-2}$

16 b)

$15 + 18 + 36 = 69 \text{ m}$

16 c)

$25 + 12.5 = 37.5 \text{ m}$

16 d)

69 m

16 e)

$69 - 37.5 = 31.5 \text{ m}$

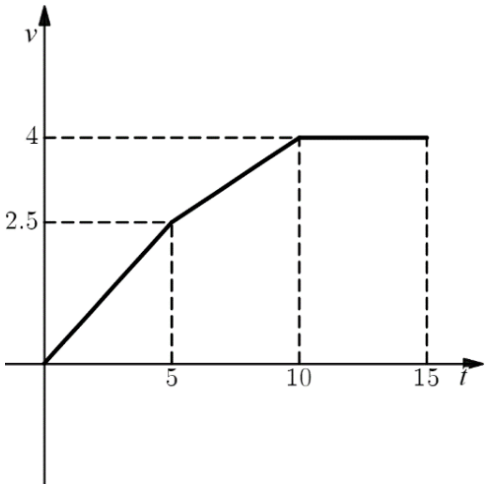
16 f)

$69 + 37.5 = 106.5 \text{ m}$

16 g)

$\frac{106.5}{35} = 3.04 \text{ ms}^{-1}$ (3 SF)

17 a)



17 b)

4 ms^{-1}

17 c)

$\frac{1}{2} \times 5 \times 2.5 + \frac{1}{2}(2.5 + 4) \times 5 + 20 = 42.5 \text{ m}$

18 a)

$0\ \text{ms}^{-2}$, $-\frac{21}{7} = -3\ \text{ms}^{-2}$, $\frac{6}{12} = 0.5\ \text{ms}^{-2}$

18 b)

On line $v = -3t + 30$ so $t = 10$

$t = 10$ s and $t = 24$ s

18 c)

$\frac{1}{2}(10 + 5) \times 15 + \frac{1}{2} \times 14 \times 6 = 112.5 + 42$

$= 154.5$ m

18 d)

$112.5 - 42 = 70.5$ m

19 a)

$-\frac{25}{5}$ so max acceleration is $5\ \text{ms}^{-2}$

19 b)

On line $v = -5t + 62$ so $t = 12.4$

$t = 0$ s, $t = 12.4$ s and $t = 35$ s

19 c)

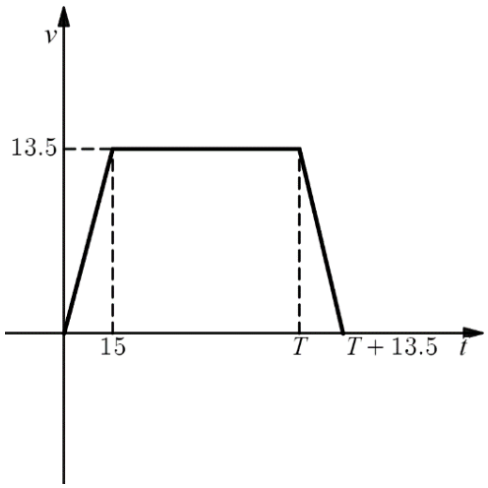
$\frac{1}{2}(4 + 12.4) \times 12 - \frac{1}{2}(3 + 22.6) \times 13 = -68$ m

So 68 m before the initial start point.

20 a)

$15 \times 0.9 = 13.5\ \text{ms}^{-1}$

20 b)



20 c)

$\frac{1}{2}(T + 13.5 + (T - 15)) \times 13.5 = 918$

$T + 13.5 + T - 15 = 136 \Rightarrow 2T = \frac{275}{2}$

$\Rightarrow T = 68.75$

So total time is $68.75 + 13.5 = 82.25$ s

21 a)

$\frac{1}{2} \times t_1 \times 6 = 15$ so $t_1 = 5$ s

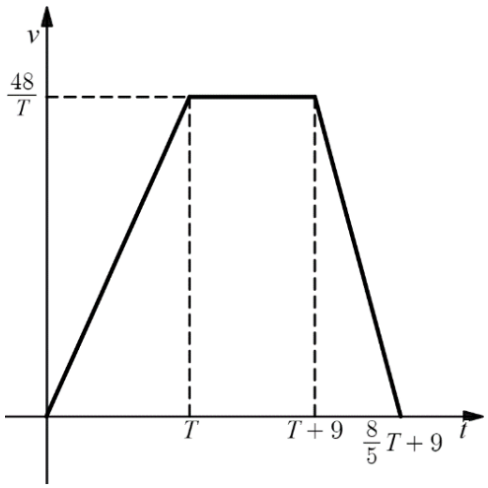
$(t_2 - 5) \times 6 = 200 - 15$ so $t_2 = \frac{215}{6}$

so $t = 35\frac{1}{6}$ seconds

21 b)

$\frac{1}{2} \times 3 \times 6 = 9$ so 209 metres

22 a)



22 b)

$\frac{1}{2}\left(\frac{8}{5}T + 9 + 9\right) \times \frac{48}{T} = 146.4$

$\Rightarrow \frac{8}{5}T + 9 + 9 = 6.1T \Rightarrow 18 = 4.5T \Rightarrow T = 4$

23 a)

$\frac{8}{32} = 0.25$ ms^{-2}

23 b)

$v = 0.25t + 12$

23 c)

Overtake at time t when they have travelled the same distance:

As areas must be equal, $t < 32$ giving

$15t$ and $\frac{1}{2}(12 + 12 + 0.25t) \times t = 12t + 0.125t^2$

$15t = 12t + 0.125t^2 \Rightarrow t = 0,\ t = 24$

so at 24 s

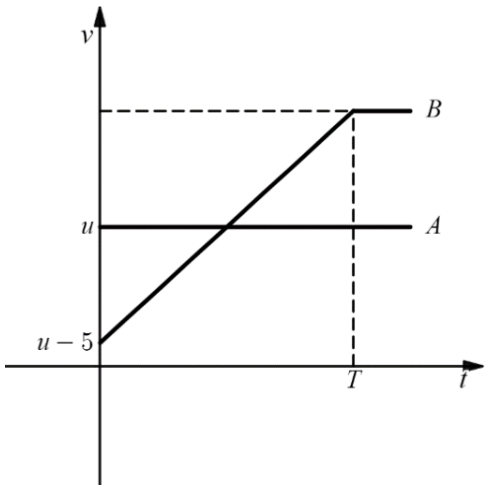
23 d)

$15 \times 24 = 360$ m

23 e)

The time at which both cars are instantaneously driving at the same speed of 15 ms^{-1}

24 a)



24 b)

Overtake at time T when they have travelled the same distance, so by symmetry $v = (u + 5)$ ms^{-1}

24 c)

$T = 2 \times 20 = 40$

$u \times 40 = 1000 \Rightarrow u = 25$ ms^{-1}